

UNIVERSITY OF BOTSWANA

DEPARTMENT OF MATHEMATICS

2017/2018 – EXAMINATIONS

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COURSE NUMBER MAT111 DURATION 2 hrs DATE NOV 2017

TITLE OF PAPER INTRODUCTORY MATHEMATICS I

SUBJECT MATHEMATICS TITLE OF EXAMINATION BSc I

MORNING/AFTERNOON

INSTRUCTIONS:

- ANSWER **ALL** QUESTIONS IN SECTION A AND ANY **TWO** QUESTIONS FROM SECTION B.
- ALL MARKS ARE INDICATED IN [].

NUMBER OF PAGES: 5

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TO DO SO BY THE SUPERVISOR.



SECTION A

Answer All Questions: Each Question carries 10 Marks.

Question 1

(a) Rationalize $\frac{1}{\sqrt{x}+1}$. [4]

(b) Factor the expression $x^3 - 2x^2 - 4x + 8$ completely. [6]

Question 2

(a) Simplify the expression $2x(1-2x)^{-\frac{5}{2}} + 3(1-2x)^{-\frac{3}{2}}$. [5]

(b) Simplify $\frac{x^3-8}{x^2-4} \div \frac{x^2+2x+4}{x^3+8}$. [5]

Question 3

(a) Sketch the graph of $f(x) = (x-2)^2 - 3$. [6]

(b) Given $f(x) = \sqrt{x+2}$ and $g(x) = \sqrt{x-2}$, find the domain of

$$h(x) = \left(\frac{f}{g}\right)(x).$$

[4]

Question 4

Given that $f(x) = 3x + 2$.

- (a) Show that f is one to one. [4]
- (b) Find the f^{-1} , the inverse of f . [3]
- (c) Find $f \circ f^{-1}$. [3]

Question 5

- (a) (i) Determine which quadrant the angle $\theta = \frac{5\pi}{6}$ is in. [2]
- (ii) Evaluate $\sin \frac{5\pi}{6}$ without the use of a calculator. [2]
- (iii) Evaluate $\cot \frac{5\pi}{6}$ without the use of a calculator. [2]
- (b) Verify the identity
 $(\sin x + \cos x)^2 = 1 + \sin 2x$. [4]

Question 6

- (a) Write $5 \log(x+2) - \frac{1}{2} \log(2x-1) + \log x^2$ as a single logarithm. [5]
- (b) Solve $\log_3(5x)^3 = 12$ [5]

SECTION B

Answer TWO Questions OUT of FOUR:

Each Question carries 20 Marks

Question 7

- (a) Determine whether the function below is odd, even or neither.
(Show your working)

$$f(x) = x^3 - 3x.$$

[5]

- (b) For $f(x) = x^2 - 2x + 1$, find

$$\frac{f(x+h) - f(x)}{h}.$$

[6]

- (c) Verify that the functions $f(x) = x^3 + 7$ and $g(x) = (x - 7)^{\frac{1}{3}}$ are inverse of each other.

[9]

Question 8

- (a) Solve the quadratic equation below by completing the square.

$$2x^2 + 8x + 9 = 0.$$

[7]

- (b) Solve the inequality and graph the solution on the real line

$$2x^2 - 5x > -3.$$

[7]

- (c) Write the complex number $\frac{2}{1+i} - \frac{3}{1-i}$ in standard form.

[6]

Question 9

- (a) (i) Give the angle 150° in radians and $\frac{\pi}{6}$ in degrees. [2]
(ii) Find the exact value of $\cos(\frac{21\pi}{4})$. [4]
- (b) Find all the solution of the equation below in the interval $[0, 2\pi]$.
 $4 \cos^2 x - 3 = 0$. [4]
- (c) Let $f(x) = \cos x$ and $g(x) = -2 \cos(x - \pi/2) + 1$.
(i) Describe the sequence of transformation that can be applied to the graph of f to obtain the graph of g . [6]
(ii) Find the amplitude and period of g . [2]

Question 10

- (a) Solve $2^{x+1} - 3(2^{-x}) = -5$. [7]
- (b) Given the function $f(x) = x^3 - 9x$,
(i) find the x intercepts and the y intercept, [4]
(ii) use the leading term to describe the left hand and the right hand behaviour of the graph of f , hence sketch the graph of f . [6]
- (c) Find the quotient and the remainder by dividing out
$$\frac{2x^3 + 3x^2 - 5}{x^2 + 2}$$
. [3]